GENERAL CONCEPTS OF MUSCLES

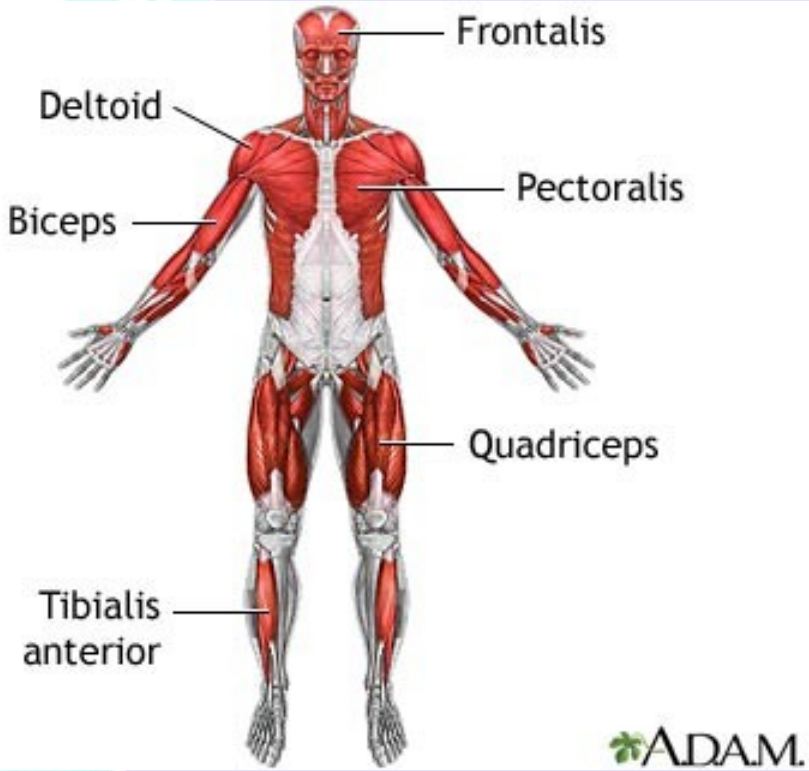
LEARNING OBJECTIVES

- At the end of the lecture the student should be able to:
- Enlist the components of muscular system.
- Classify the muscles according to:
  - Their shape (architecture) and
  - Direction of fibres.
  - Their action.
- Express the principles of innervation and blood supply of muscles.

MUSCLES

- **Muscle** (from Latin "mouse") is the contractile tissue which produces movements.

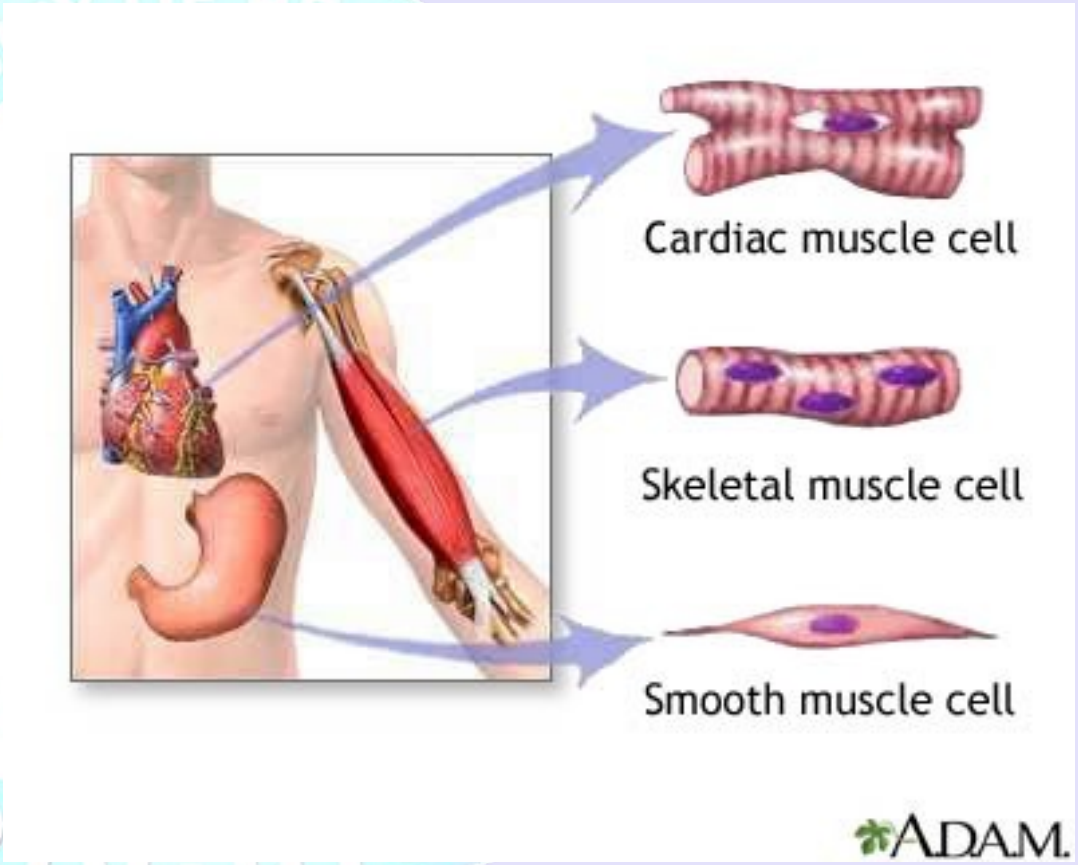
- Derived from the mesodermal layer of embryonic germ cells, but some from the ectoderm (e.g. muscles of the iris).



CLASSIFICATION OF MUSCLES

Morphological classification (based on structure)

1. **Striated**
   - ❖ Skeletal muscle
   - ❖ Cardiac muscle
2. **Non striated or smooth**
   - ❖ Smooth muscle

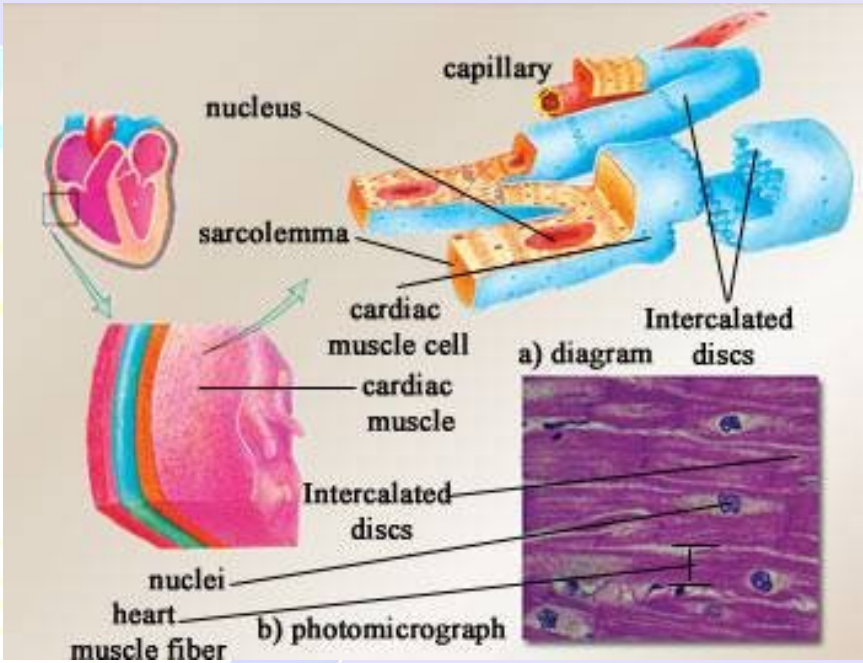


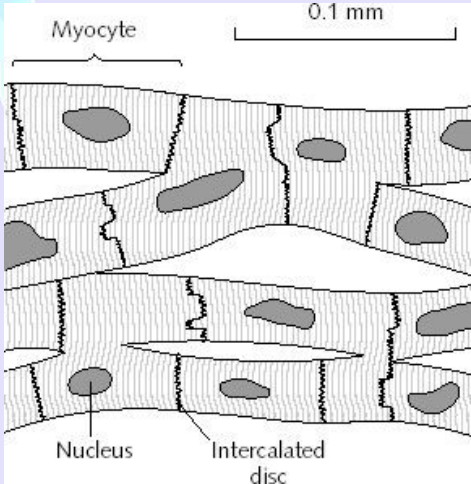
Functional classification

1. **Voluntary**
   - ❖ Skeletal muscle
2. **Involuntary**
   - ❖ Cardiac muscle
   - ❖ Smooth muscle

CARDIAC MUSCLE

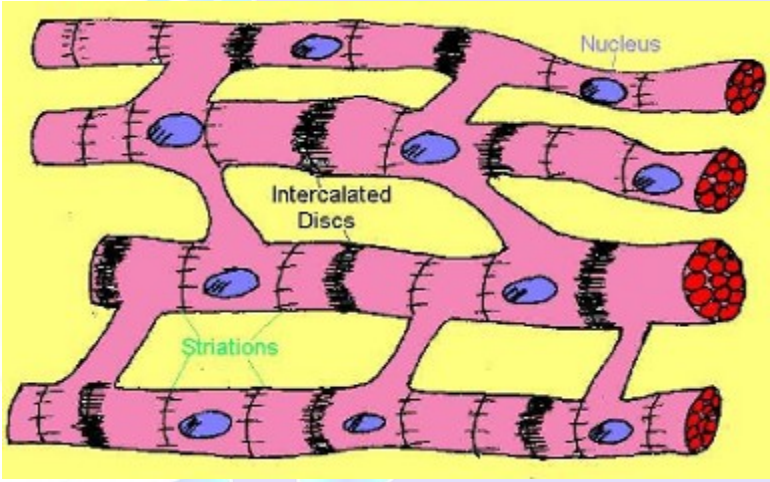
- Cardiac muscle consists of muscle cells with one centrally placed oval nucleus.
- Cardiac muscle is innervated by the autonomic nervous system.
- Cardiac muscle exhibits cross-striations.
- Cardiac muscle is also called involuntary striated muscle.





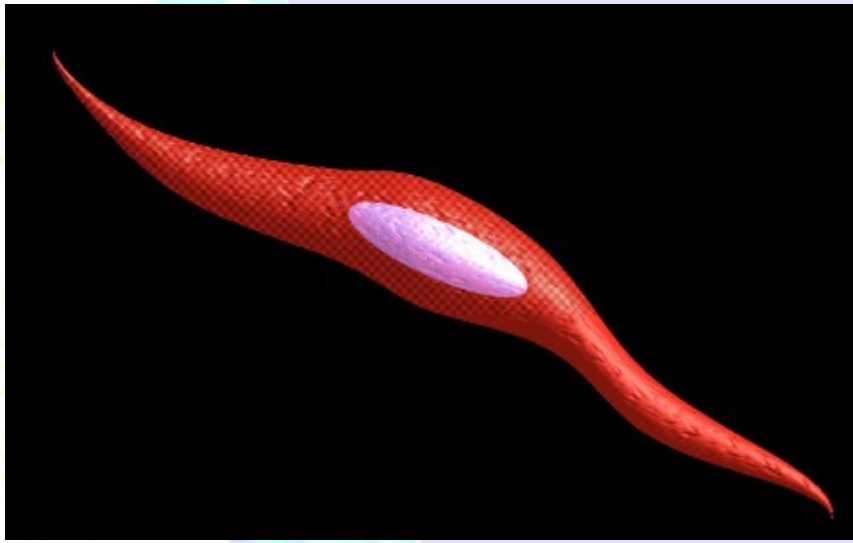
CARDIAC MUSCLE

- Cross-striations in cardiac muscle are less distinct than skeletal muscle.
- In contrast to skeletal muscle cells, cardiac muscle cells often branch at acute angles
- They are connected to each other by specializations of the cell membrane in the region of the **intercalated discs.**
- Intercalated discs invariably occur at the ends of cardiac muscle cells



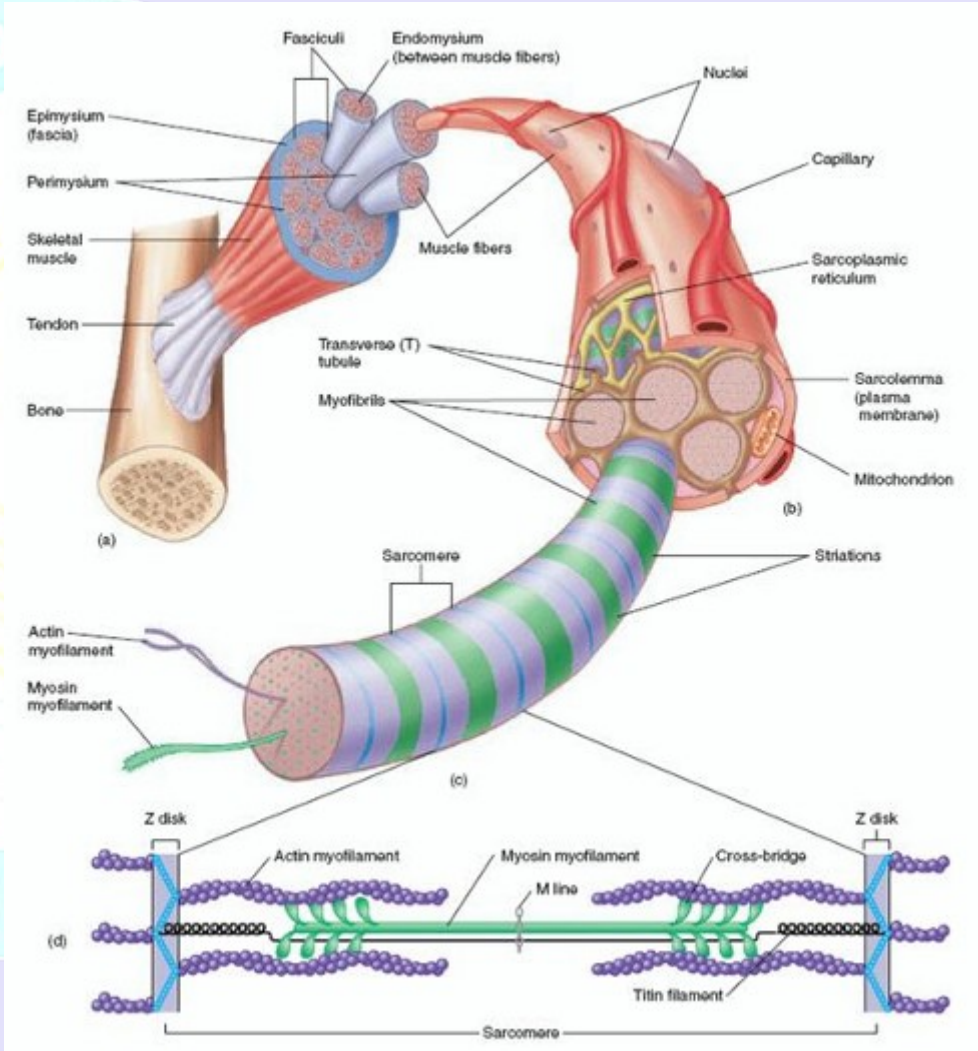
SMOOTH MUSCLE

- They are so called because they don’t exhibit cross striations on light microscopy.
- Smooth muscle consists of spindle shaped cells of variable size.
- The largest smooth muscle cells occur in the uterus during pregnancy.(500 micron).The smallest are found around small arterioles.
- Smooth muscle cells contain one centrally placed nucleus.
- The innervations of smooth muscle is provided by the autonomic nervous system.
- Smooth muscle makes up the visceral or involuntary muscle.



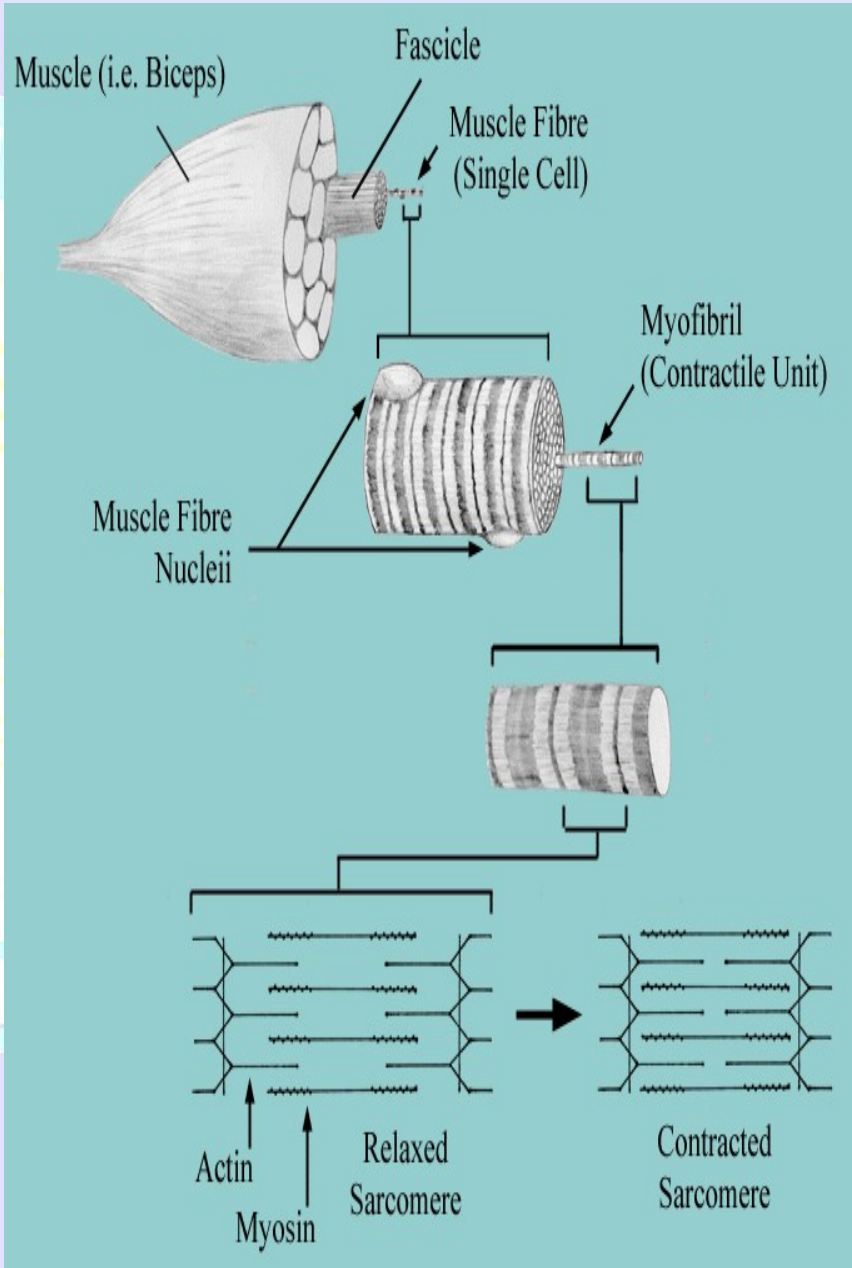
SKELETAL MUSCLE

- Most abundantly attached to Skeleton.
- Showing cross striation under the microscope.
- Supplied by somatic nerves, therefore under the voluntary control.
- Respond quickly to stimuli, producing rapid contraction, that’s why fatigue quickly.
- They are the motor of the body.



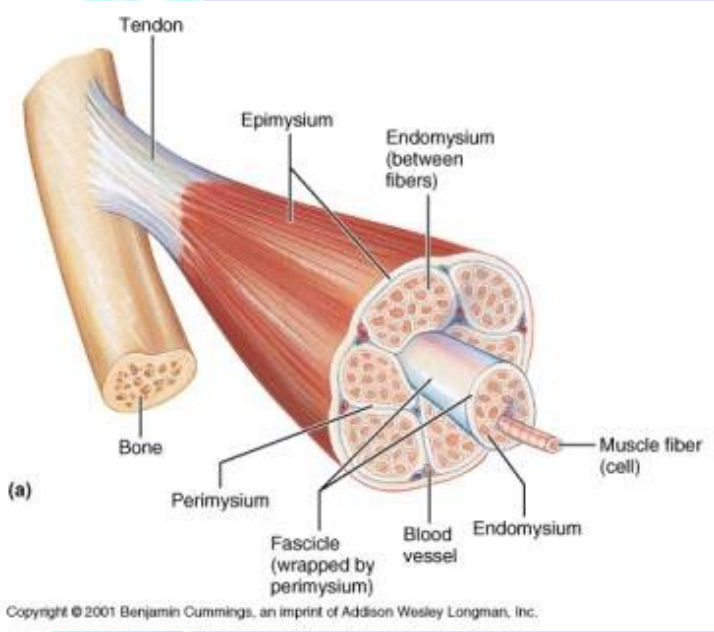
SKELETAL MUSCLE

- Skeletal muscle consists of very long tubular cells (also called muscle fibers).
- Skeletal muscle fibers contain many peripherally placed nuclei.
- Skeletal muscle fibers show in many preparations characteristic cross-striations. It is therefore also called striated muscle.
- Skeletal muscle is innervated by the somatic nervous system.
- Skeletal muscle makes up the voluntary muscle.



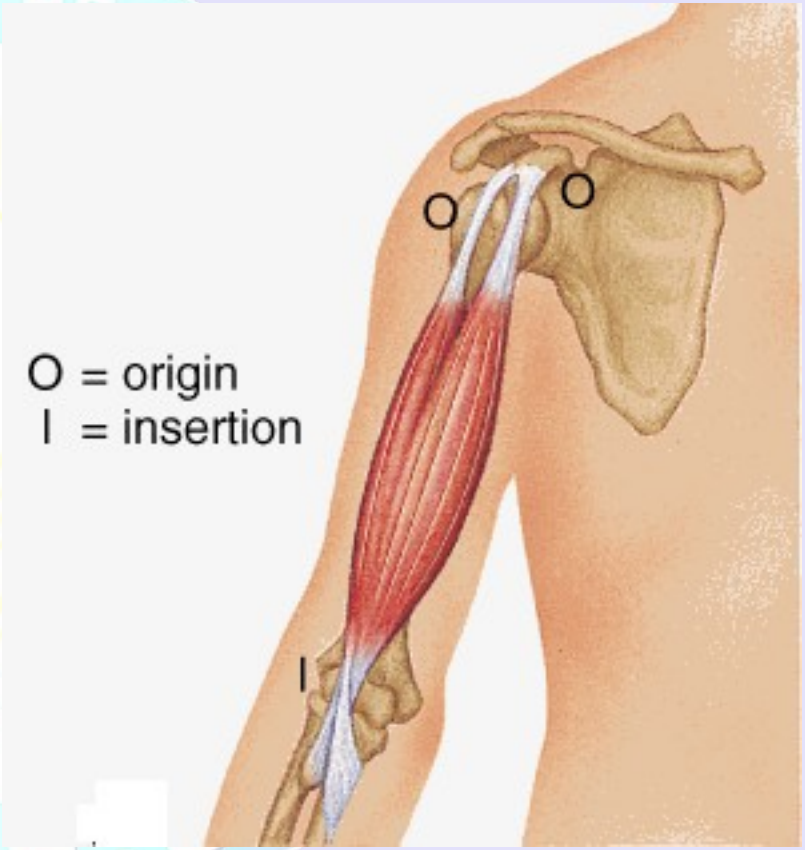
COVERINGS OF MUSCLE

- **Epimysium.**
  - Entire muscle is surrounded by a connective tissue called the *Epimysium*.
- **Perimysium**.
  - Fascicles are surrounded by a connective tissue sheath called the perimysium.
  - Fascicles are actually bundles of individual muscle cells or muscle fibers.
- **Endomysium.**
  - Each muscle cell or muscle fiber is surrounded by a connective tissue sheath known as the endomysium.
  - Each fascicle is made up of several muscle cells known as muscle fibers.



PARTS OF SKELETAL MUSCLE

- **Origin**
  - The one end of muscle which remains fixed during contraction (Proximal End usually)
- **Insertion**
  - The other end which moves more during contraction (Distal End usually)
- **Belly**
  - Central fleshy contractile part.
- **Tendon or aponeurosis**
  - Fibrous non-contractile part. When it is cord like called as tendon or when broad sheath like, aponeurosis.



Structural Classification of Skeletal Muscle

PARALLEL MUSCLE

- The muscle fasciculi are parallel to the line of pull
- Consists of parallel fibers all of equal size
- The range of movement is maximum
- The force of contraction is less
- When muscle contracting all the muscle fibers are contacting with equal length

Structural Classification of Skeletal Muscle

On the basis of arrangement of fibers

A- Parallel

1- Quadrilateral

2- Rhomboid

3- Strap like

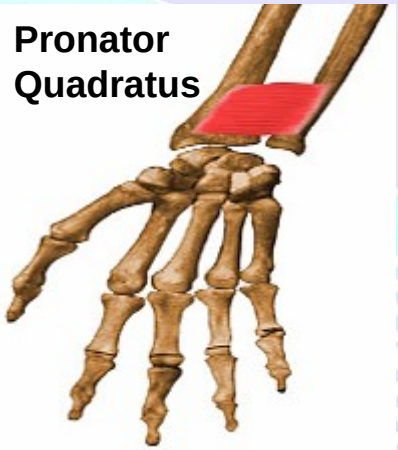
4-Strap like with tendinous intersections(rectus abdominis)

5- Fusiform

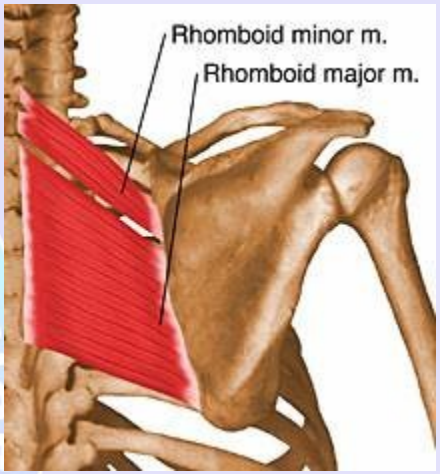
Quadrilateral



Pronator Quadratus

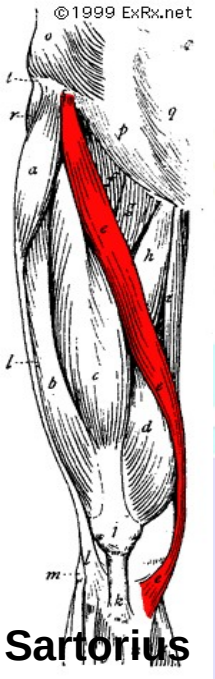


Rhomboidius Major & Minor



Strap like



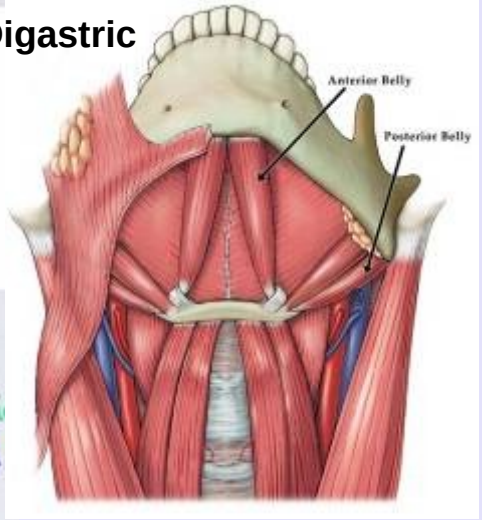


Sartorius

Fusiform



Digastric



Structural Classification of Skeletal Muscle

OBLIQUE MUSCLE

- The muscle fasciculi are oblique to the line of pull
- Consists of oblique fibers (feather like) muscle fibers are variable size (some are shorter while other are loner)
- The range of movement is reduced
- The force of contraction is powerful
- When muscle contracting all the muscle fibers are contacting with unequal length

Structural Classification of Skeletal Muscle

On the basis of arrangement of fibers

B- Oblique

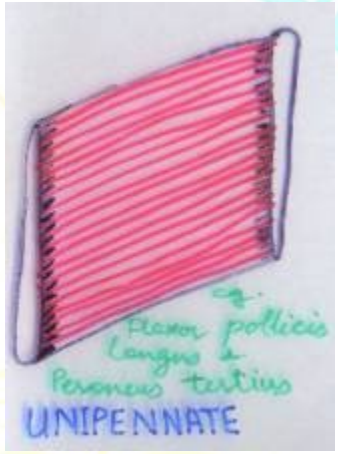
1-Unipennate

2- Bipennate

3- Multipennate

4- Circumpennate

Unipennate



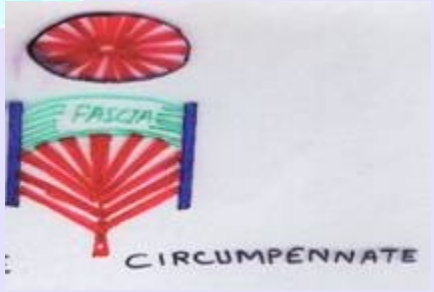
Bipennate



Multipennate



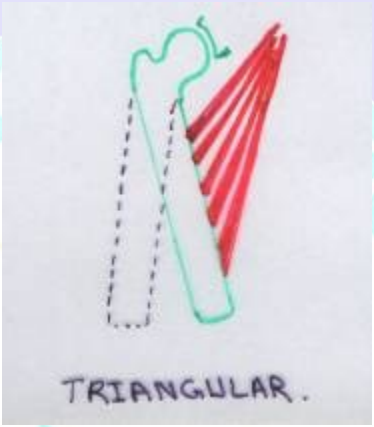
Circumpennate



Classification of Skeletal Muscle

5- Triangular

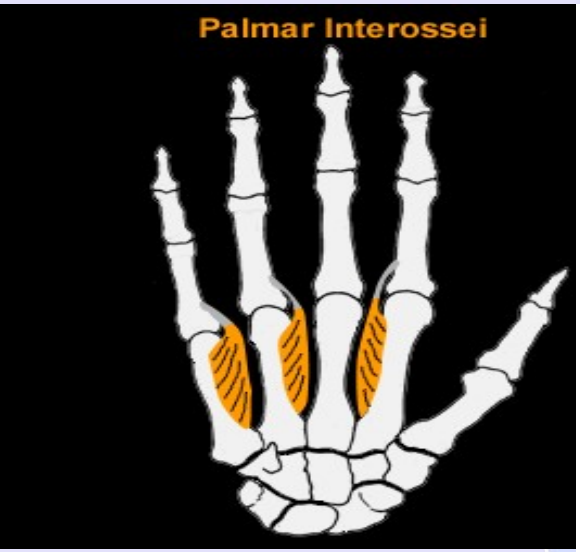
6- Spiral or Twisted



Unipennate

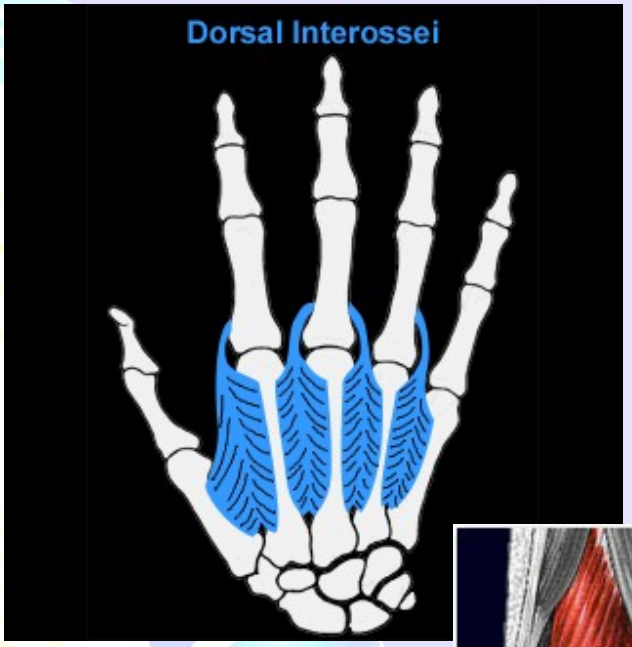
Bipennate

Palmar Interossei



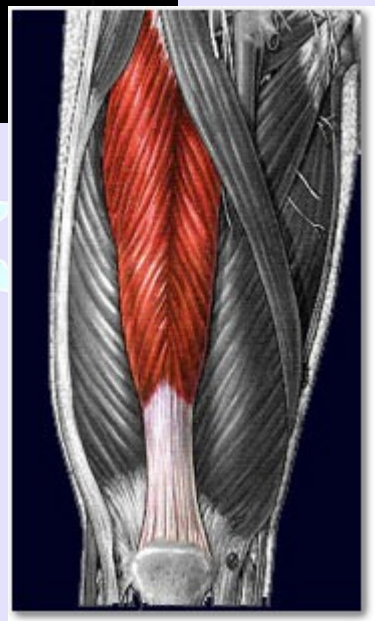


Dorsal interossei



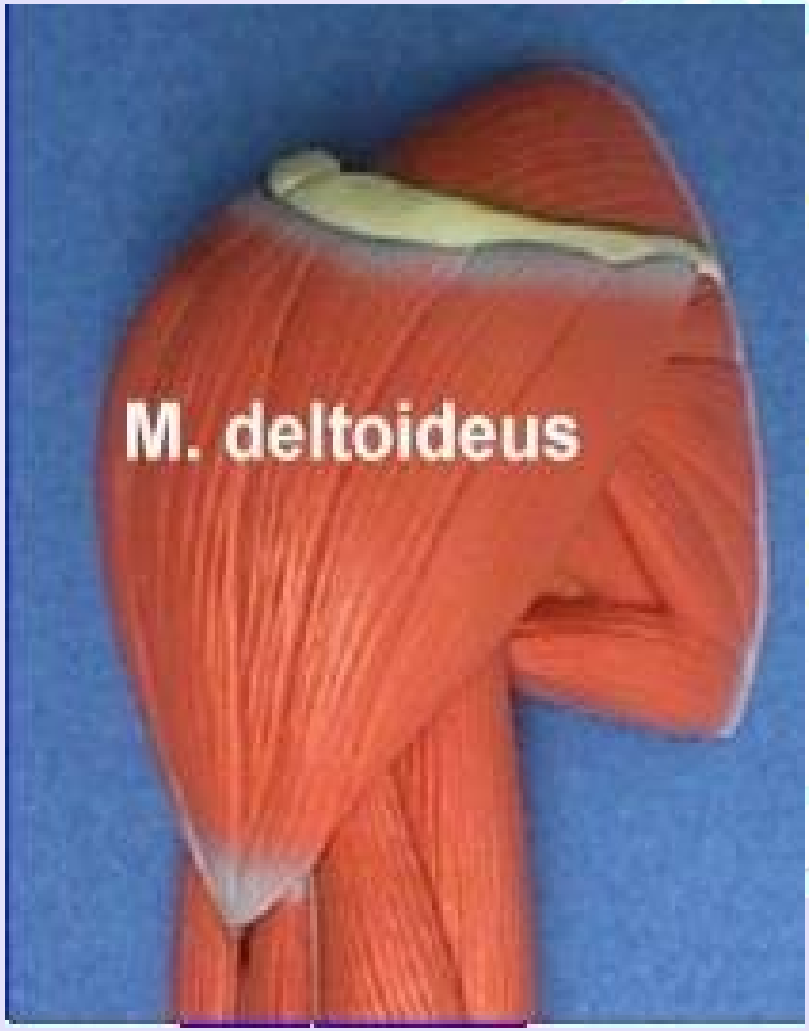
Flexor Pollicis Longus

Rectus Femoris Muscle



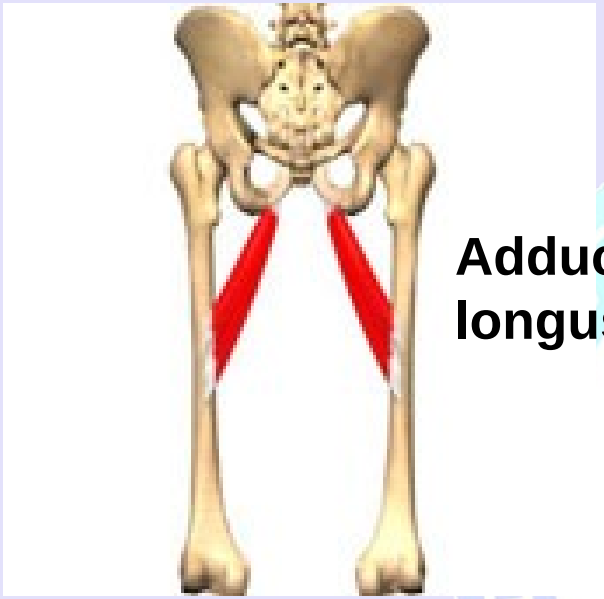
Multipennate

Subscapularis



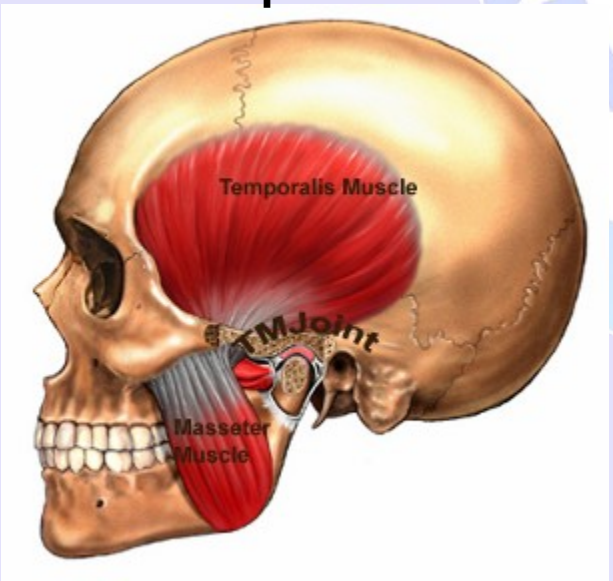


Triangular



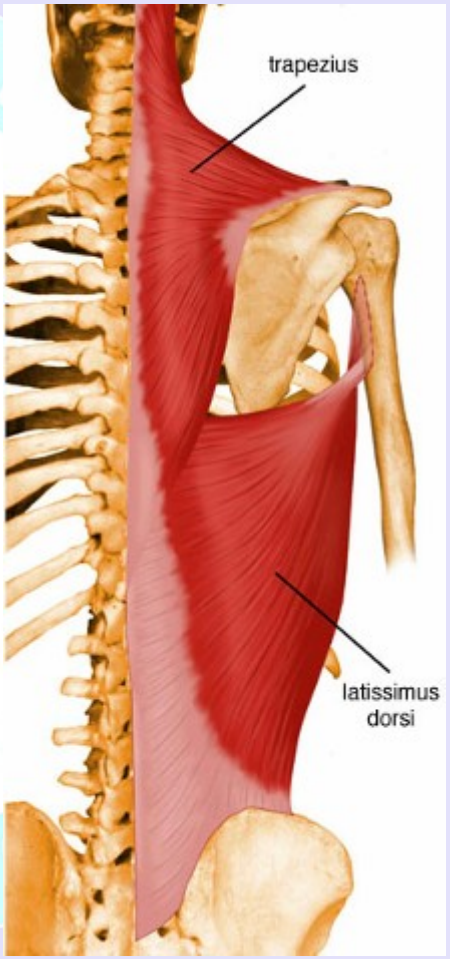
Adductor longus

Temporalis



Spiral or Twisted

Latissimus dorsi





major & minor.

RHOMBOID



eg. Thyrohyoid.

QUADRI - LATERAL



eg. Sartorius & Sternothyroid

STRAP



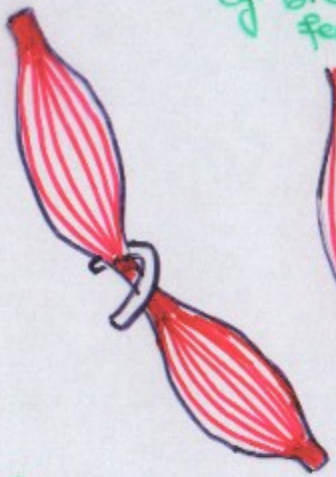
eg. Rectus Abdominis.

STRAP WITH TENDINOUS INTERSECTIONS



eg. Digastric

FUSIFORM.



eg. Bicep femoris



TWO HEADED

TWO BELLIES

eg. Omohyoid.



TRIANGU - LAR.

eg. Temporalis & Add. longus.



eg. Flexor pollicis longus & Peroneus tertius

UNIPENNATE



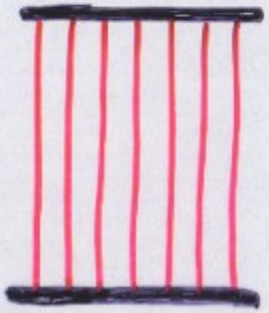
eg. Rectus femoris & Peroneus longas

BIPENNATE



eg. Acromial fibre of Deltoid

MULTIPENNATE



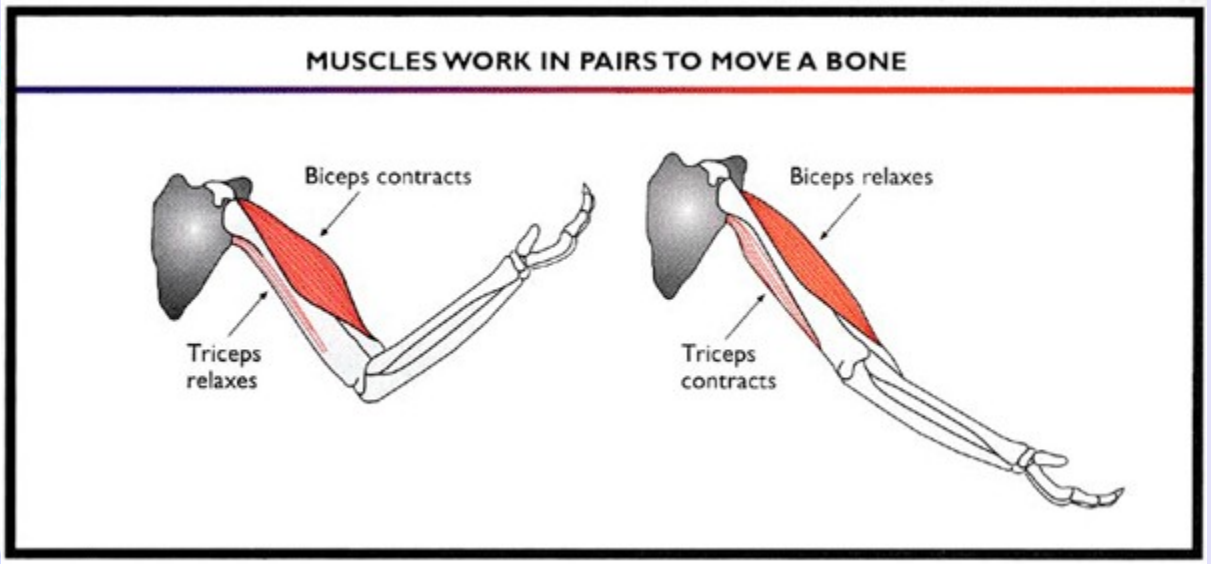
RELAXED



CONTRACTED.

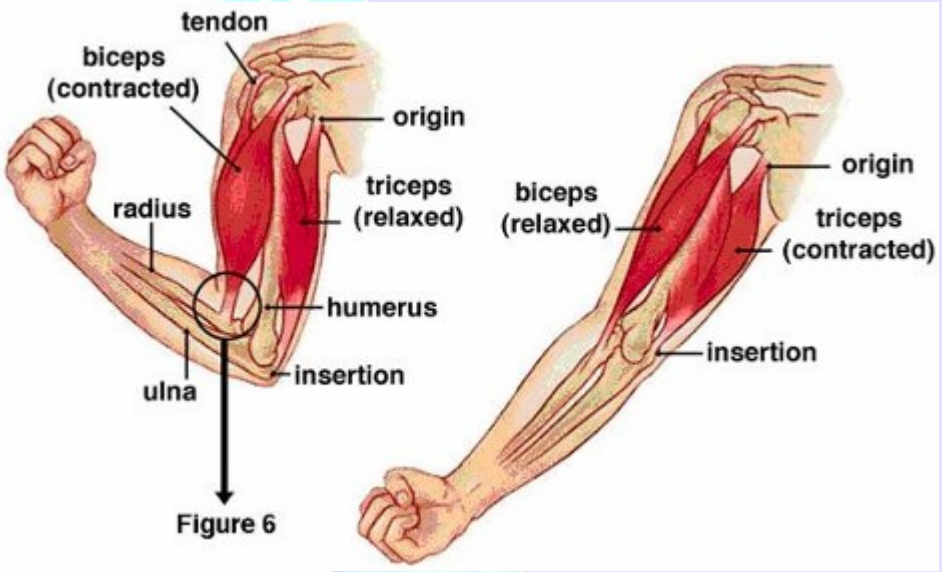
FUNCTIONAL CLASSIFICATION OF MUSCLE

- Agonist
- Antagonist
- Synergist
- Fixator



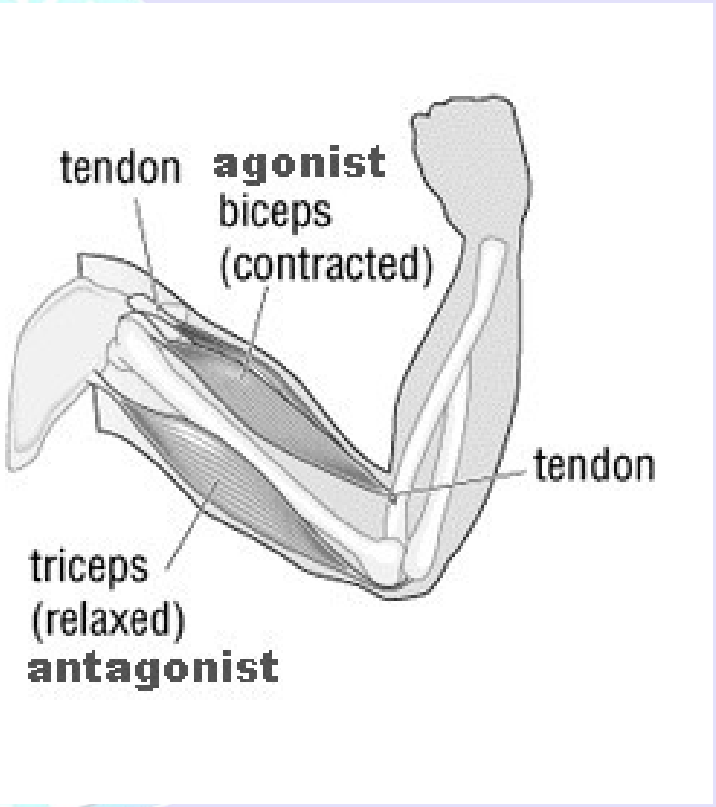
AGONIST (MUSCLE)

- Agonist is kind of muscle that causes movement to occur. It creates the normal range of movement in a joint by contracting.
- Agonists are also referred to as "prime movers" since they are the muscles that are primarily responsible for generating movement.



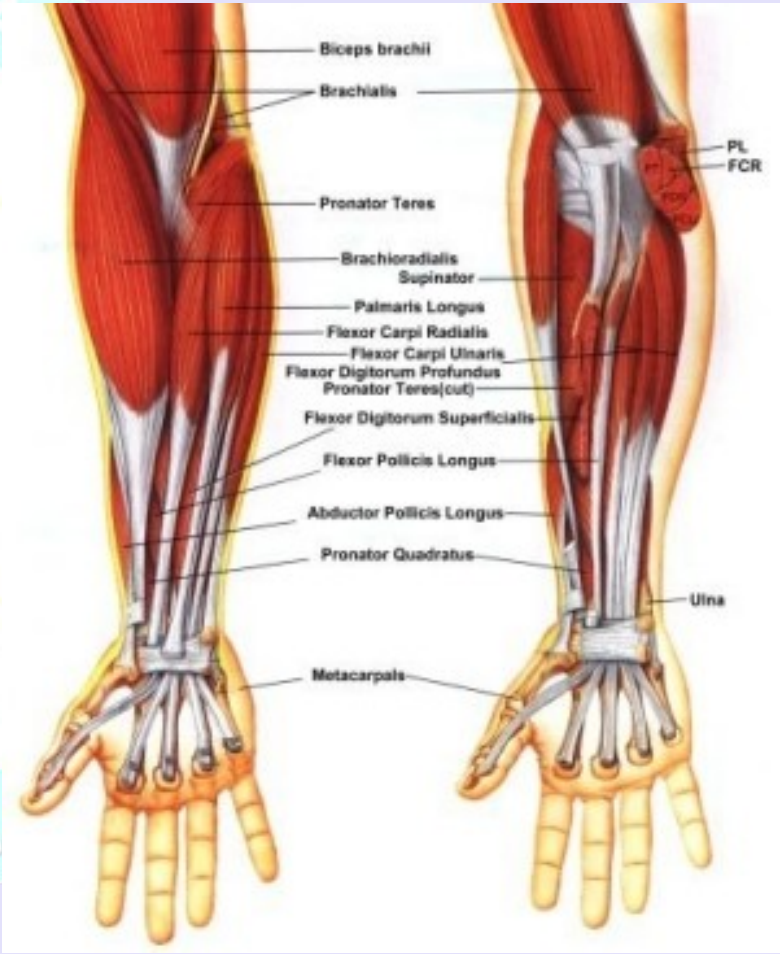
ANTAGONIST

- An **antagonist** is a kind of muscle that acts in opposition to the movement generated by the agonist and is responsible for returning a limb to its initial position.



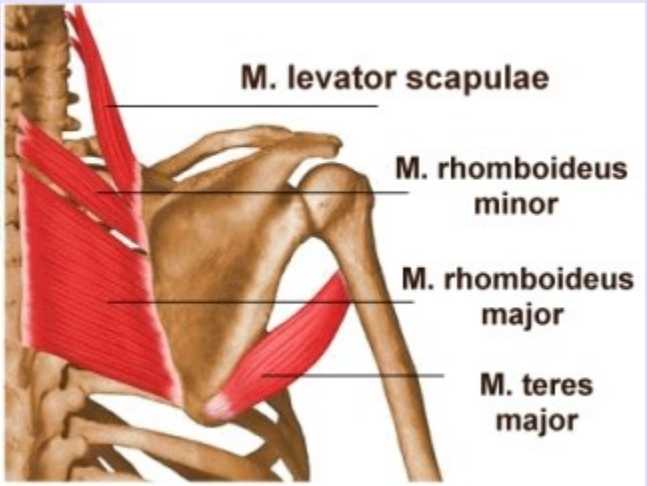
SYNERGIST

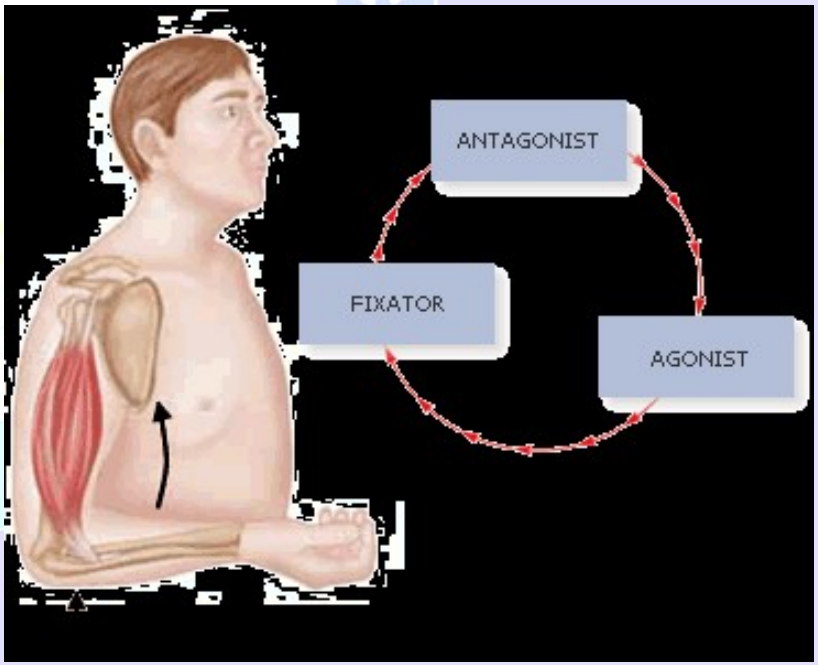
- Synergist complements the action of prime movers.
- **Synergist** is a kind of muscle which perform, or assist in performing, the same set of joint motion as the agonists.



FIXATOR MUSCLE

- A muscle that acts as a stabilizer of one part of the body during movement of another part.
- **Fixator** - a fixator holds the origin of the agonist, stable. E.g., abducting arm - deltoid is abductor, fixators are other flexors keeping the scapula in position e.g. rhomboids.



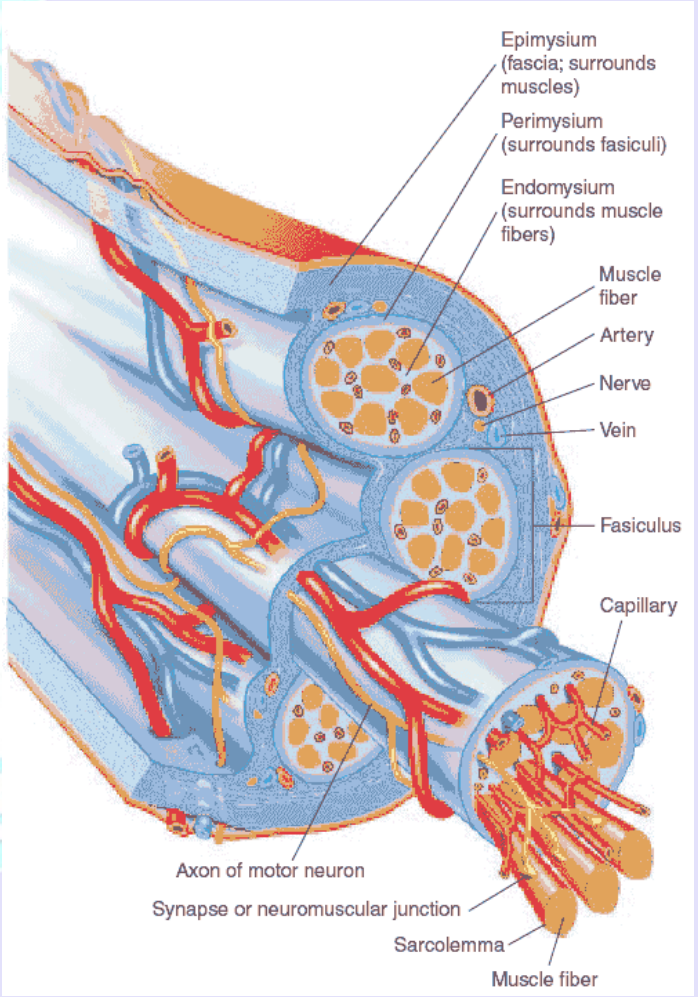


BLOOD SUPPLY AND NERVE SUPPLY OF MUSCLE

- Arteries supply the structure to which the muscles are in contact (neighbouring vessels)

Nerve supply:

- Motor nerves enter the fleshy portion of muscle, almost always from the deep portion called motor point.



REFERENCES

- Gray’s human anatomy.
- KLM clinical oriented anatomy



Thank You